AI Usage Note

Runbook Following Agent

Project Overview

The Runbook Following Agent is an AI-powered operational assistant designed to automate incident response by interpreting structured Markdown runbooks and guiding users through troubleshooting workflows. The system analyzes incidents, retrieves relevant runbooks, executes logical steps, validates outcomes, and generates resolution reports, thereby reducing manual effort and improving operational consistency.

AI Tools Used

ChatGPT (OpenAI) and Claude (Anthropic) were used throughout the development lifecycle to accelerate design, implementation, debugging, testing, and documentation activities.

How AI Assisted Development

AI was used as a development accelerator rather than a replacement for engineering judgment. It assisted in system architecture design, workflow modeling, Markdown runbook parsing strategies, incident-analysis logic, Streamlit interface development, Python code generation, report generation, testing strategies, GitHub documentation, and deployment support.

Key Engineering Contributions Enabled by AI

• Design of the Runbook Retrieval and Execution workflow. • Development of the step-by-step incident resolution engine. • Creation of Markdown-based runbook structures. • Generation of sample operational runbooks for infrastructure incidents. • Streamlit dashboard implementation. • Error handling and validation logic. • Incident report generation and output formatting. • Testing and debugging support.

What AI Got Wrong

AI-generated outputs occasionally contained incorrect troubleshooting sequences, incomplete validation logic, dependency conflicts, indentation issues, and assumptions that were unsuitable for production workflows. Several generated commands required manual verification before integration.

Manual Engineering Work Performed

Significant manual effort was invested in validating and refining all AI-generated outputs. Manual contributions included runbook design validation, workflow optimization, debugging syntax errors, improving parsing accuracy, implementing execution safeguards, enhancing user experience, testing incident scenarios, and verifying command correctness. No AI-generated code was used without review and modification.

Representative Prompts Used

1. Design a Runbook Following Agent capable of reading Markdown runbooks and executing troubleshooting workflows. 2. Create an incident-analysis workflow that identifies the next operational action based on runbook instructions. 3. Generate structured runbooks for database failures, API outages, high CPU utilization, and disk-space incidents. 4. Build a Markdown parser that extracts steps, commands, conditions, expected outputs, and escalation rules. 5. Design an execution engine that validates step completion before proceeding. 6. Generate incident-resolution reports summarizing actions performed and outcomes achieved. 7. Debug Streamlit runtime issues and improve application stability.

Impact of AI on Development

AI significantly reduced development time by accelerating ideation, code prototyping, debugging, documentation, and workflow design. Tasks that would normally require extensive research and experimentation were completed more efficiently, allowing greater focus on validation, testing, and solution refinement.

Declaration

All AI-generated content was carefully reviewed, tested, modified, and integrated manually. Final architectural decisions, implementation choices, validation logic, testing activities, and deployment configurations were performed by the project team. AI served as an assistive tool, while responsibility for correctness, reliability, and project outcomes remained entirely with the developers.

Conclusion

The Runbook Following Agent demonstrates the effective use of Generative AI as a software engineering assistant. AI accelerated development and improved productivity, while human oversight ensured technical accuracy, reliability, and alignment with project objectives. The final system reflects a combination of AI-assisted development and substantial manual engineering effort.